Hemanjali Reddy

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EXPERIENCE

Pfizer Inc. **New York, NY**
AI Engineer *Jun 2026 – Aug 2026*

- Architected a **system-agnostic evaluation framework** for agentic AI combining LLM-as-a-Judge, trajectory analysis and drift detection, applied to a **clinical-trial oncology assistant built on Google ADK**.
- Designed a **context-aware judge layer** scoring responses against retrieved data on a **4-dimension rubric**, using **score distributions across runs** to benchmark LLMs and test judge reliability.
- Engineered **perturbation-based robustness tests** that surfaced failure modes accuracy checks miss, including a paraphrase case where **answer similarity fell to 0.13**.
- Built **latent-space trajectory analysis** embedding each step from query to final answer to pinpoint **where repeated and perturbed runs diverged**, with UMAP clustering for answer consistency.

ResMed Technologies **Bengaluru, India**
Associate Machine Learning Engineer *Jul 2024 – Jul 2025*

- Built the **agent harness** for a production **patient-facing medical chatbot** with LangGraph, served as containerized FastAPI services on AWS, letting discharged patients ask about diagnoses and medications, **easing nurse workload** across **15.9M+ requests to date**.
- Engineered the chatbot's context layer and **agent scaffolding**, integrating customer-specific instructions, clinical context, retrieval, safety constraints for accurate, safe answers.
- Built an **automated regression-evaluation suite** that benchmarks every prompt, model, context and harness change across quality, latency, safety and failure modes, guiding prompt engineering toward the best-performing prompts and **reducing experiment runtime.**
- Created **LangSmith** and **Datadog** observability dashboards tracking agent traces, AWS service utilization and service latency, **surfacing bottlenecks** that guided latency optimization.
- Led the migration from Guardrails AI to **AWS Bedrock Guardrails**, **cutting end-to-end latency by 37% (19s to 12s)** while strengthening privacy, safety and HIPAA-aligned controls.
- Built an **agentic RAG system** for insurance claims that routes each query to a SQL agent over claims data or vector retrieval over HME/DME documentation, unifying **structured and unstructured sources** in a single LLM-synthesized answer.
- Benchmarked models across **20M+ insurance records** and selected/optimized CatBoost via cross-validation, hyperparameter tuning, and drift analysis to improve classification performance.

Mobileum Technologies **Bengaluru, India**
Machine Learning Intern *Jan 2023 – Jun 2023*

- Trained and benchmarked Decision Tree, Random Forest, and deep learning model for spam-call detection using telecom Call Detail Records, comparing classification performance across approaches.

RESEARCH & TEACHING EXPERIENCE

Cornell University **Ithaca, NY**
Graduate Research Assistant — Comparative Constitutional Analysis with LLMs *Feb 2026 – May 2026*

- Benchmarked LLMs against human experts across **581 constitutions and 460 variables**, achieving **71% agreement** and automating legal-document ingestion and inter-rater reliability.

Cornell University **Ithaca, NY**
Graduate Research Assistant — Multimodal AI for Harmful Speech Detection *Jan 2026 – May 2026*

- Benchmarked **vision-language models** on real-world, code-mixed multimedia for robustness across culturally nuanced harm categories.

Cornell University **Ithaca, NY**
Teaching Assistant — STSCI 2100: Introductory Statistics and Data Science *Aug 2025 – Dec 2025*

- Led weekly undergraduate labs in R programming, statistics and data analysis.

PROJECTS

Probabilistic Calibration of LLMs – Los Alamos National Laboratory Capstone **2025 – 2026**

- Built an LLM calibration framework comparing six prompting strategies using Brier score, ECE, and cross-entropy; found calibration error was **6–8x lower** than foundational ML baselines, with self-critique prompting recovering up to **52%** of error in the hardest setting.

InferBench — LLM Inference Performance Lab **2026**

- Built a modular CPU-based LLM benchmarking framework analyzing KV caching and CPU threading, measuring 3.4× faster decoding and 56% lower latency in tested configurations.

EDUCATION

Cornell University **Ithaca, NY**
Master of Professional Studies in Applied Statistics *CGPA: 3.9/4.0 — Aug 2025 – May 2026*

- **Relevant Coursework:** Linear Models with Matrices, Multivariate Analysis, Probability Models, Data Mining, SAS High-Performance Computing

Vellore Institute of Technology **Vellore, India**
B.Tech. & M.Tech. (Integrated), Computer Science Engineering, Minor in Data Science *CGPA: 9.18/10.0 — Sep 2020 – May 2025*

SKILLS & CERTIFICATIONS

Languages & Databases: Python, R, C++, SQL, SAS, PostgreSQL, DynamoDB, MongoDB
AI/ML Frameworks: scikit-learn, PyTorch, TensorFlow, NumPy, Hugging Face Transformers, LangChain, LangGraph, Google ADK
Frontend, Backend & APIs: React.js, HTML/CSS, FastAPI, REST APIs, Streamlit, Linux
MLOps, Cloud & DevOps: AWS, GCP, Docker, GitHub Actions, CI/CD, Datadog, Terraform
Certifications & Awards: AWS Solutions Architect Associate | Machine Learning – Stanford | Data Science – Udemy | **3rd Place, ResMed Hackathon**